Foundation Models: Modern Advances and Applications

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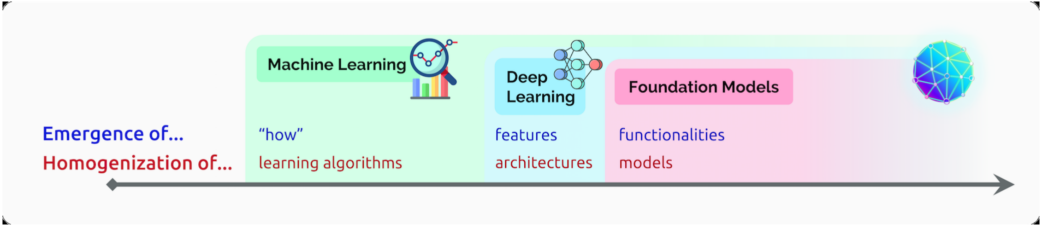
What are Foundation Models?

- **Foundation models** are large, general AI models pretrained on massive, diverse datasets.
- They provide a universal base that can be adapted (via fine-tuning or prompting) across many domains and tasks.
- The key shift: build powerful, flexible models for reuse, rather than training a new model for each task from scratch.

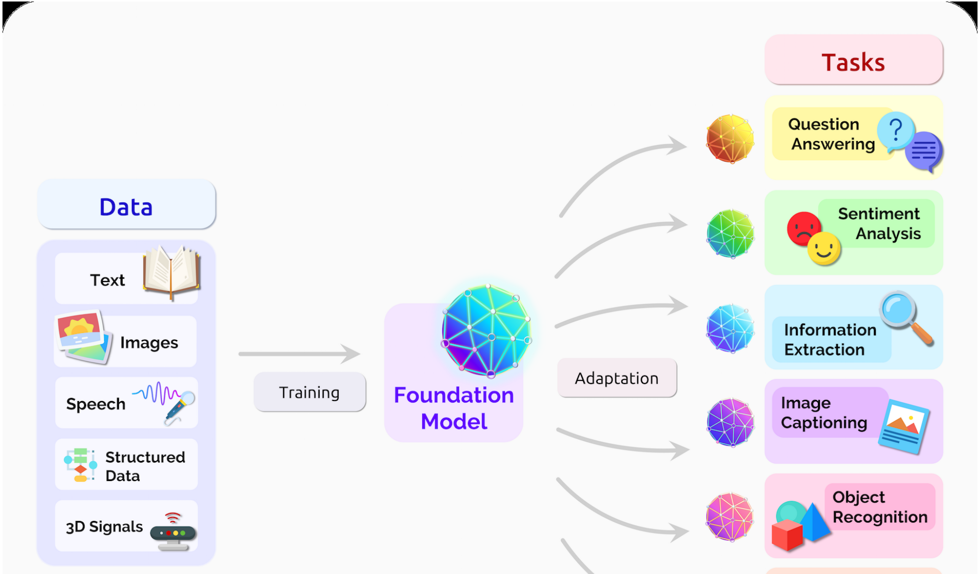


Emergence and homogenization from ML → deep learning → foundation models (Bommasani et al., arXiv:2108.07258).

Why Do Foundation Models Matter?

- **Accelerate innovation:** Dramatically reduce time and resources needed to create high-performing AI applications.
- **Expand accessibility:** Lower the expertise required for deploying AI, empowering domain experts (not just AI researchers).
- **Enable new capabilities:** Support advanced "few-shot" and "zero-shot" learning, allowing for rapid adaptation to new tasks.
- **Transform industries:** Fuel advances in medicine, science, robotics, business, and beyond.
- **Introduce new challenges:** Require careful management of data/computational resources, fairness, safety, and responsible use.

How Do Foundation Models Work?



Types of Foundation Models

- **Language models:** Trained on text data, these models can generate, summarize, and understand language.
  Examples: BERT, GPT
- **Vision models:** Trained on images/videos, they can recognize objects, segment scenes, and more.
  Examples: CLIP, SAM
- **Multi-modal models:** Learn to connect information across text, images, audio, or video.
  Examples: CLIP (text-image), Flamingo, Gemini
- **Other domains:** Foundation models exist for structured (tabular) data, medical data, code, and scientific domains.

How Do Foundation Models Work?

- **Pretraining at scale:** Train on enormous datasets (text, images, etc.) using self-supervision; learn generalized representations.
- **Adaptation:** Foundation models can be quickly adapted to new tasks using:
  - Fine-tuning on small amounts of labeled data
  - Prompting (providing instructions or context)
  - In-context learning (few-shot or zero-shot)
- **Transfer learning:** Reuse knowledge across tasks and domains, reducing the need for bespoke model training.
- **Unified architectures:** Many foundation models use architectures like transformers, enabling processing across modalities.

Examples of Foundation Models

- **BERT:** Foundation for language understanding and search.
- **GPT (Generative Pretrained Transformer):** Powers chatbots, writing assistants, and more.
- **CLIP:** Bridges vision and language—for example, "find images matching this caption."
- **SAM (Segment Anything Model):** Segments objects in any image with a single forward pass.
- **Gemini (Google):** Multi-modal model that integrates language, vision, and more.
- **PaLM, LaMDA, Imagen:** Other prominent foundation models from industry leaders.

Key Differences: Modalities, Tasks, and Adaptation

Foundation models

- Trained on **heterogeneous** data: text, images, speech, structured data, 3D/signals, . . . (see Fig. 2).
- **Homogenization:** one pretrained model is adapted to many applications; **emergence:** capabilities (e.g. in-context learning) from scale were not explicitly programmed.
- Objectives: self-supervised / contrastive / masked / alignment—not only next-token prediction.

LLMs

- **Language-first:** data are primarily text; core loss is usually autoregressive or masked LM.
- **Downstream:** generation, QA, summarization, code; multimodal **variants** pair an LLM with other encoders.
- Same "pretrain then adapt" pattern, but the **scope of pretraining data** is narrower than full multi-modal foundation models.

Implications: Ecosystem and Deployment

- **Foundation models** enable transfer across domains (vision–language, robotics, biomedicine, . . .) but concentrate capability in a few shared pretrained artifacts.
- **LLMs** excel at language and code; for vision, video, or tabular data they are usually **one component** of a larger system unless trained multi-modally from the start.
- **Limitations:** compute and data scale, inherited biases from pretraining, evaluation and safety gaps as the same "foundation" is reused widely.

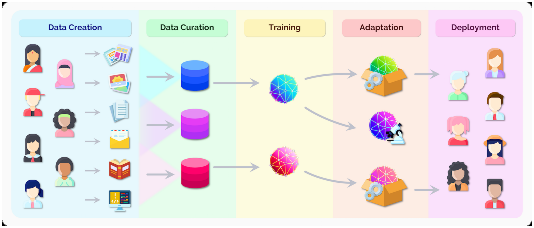
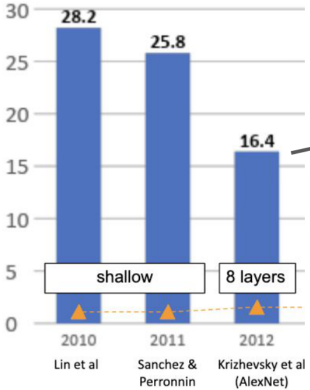
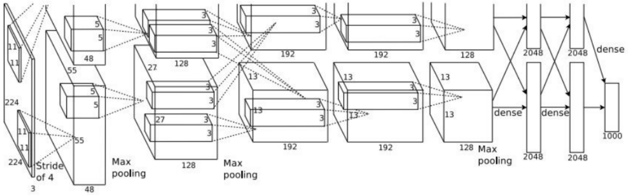


Fig. 3: From people and data through training and adaptation to deployment (Bommasani et al., arXiv:2108.07258).

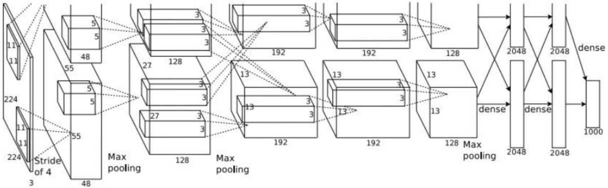




Krizhevsky et al. 2012. 8-layer “AlexNet”.

ImageNet Visual Recognition Challenge results.

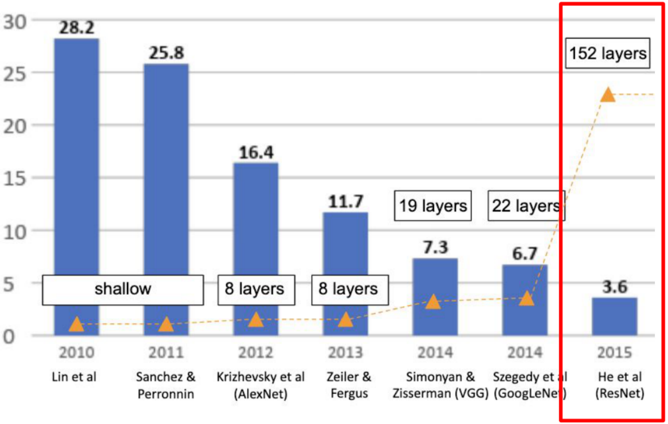
Algorithms



Compute

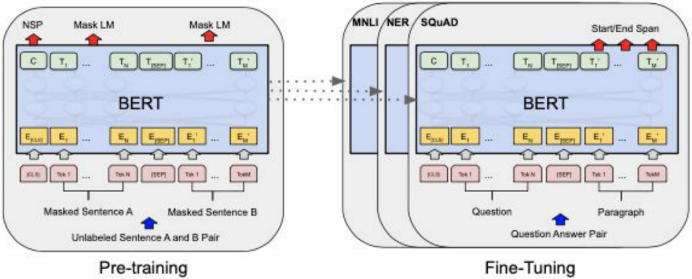
Data

2015: Very deep convnets and challenging vision tasks



He et al. 2015. ResNet.

Transformer neural network architectures (containing self-attention mechanism) and pre-training -> fine-tuning. State-of-the-art on 11 NLP benchmarks.



Vaswani et al. 2017. “Attention is all you need.”
Devlin et al. 2018. BERT.

2020: Very large-scale text generation models

SYSTEM PROMPT (HUMAN-WRITTEN)

In a shocking finding, scientist discovered a herd of unicorns living in a remote, previously unexplored valley, in the Andes Mountains. Even more surprising to the researchers was the fact that the unicorns spoke perfect English.

MODEL COMPLETION (MACHINE-WRITTEN, 10 TRIES)

```
The scientist named the population, after their distinctive horn,
Ovid's Unicorn. These four-horned, silver-white unicorns were
previously unknown to science.

Now, after almost two centuries, the mystery of what sparked this odd
phenomenon is finally solved.

Dr. Jorge Pérez, an evolutionary biologist from the University of La
Paz, and several companions, were exploring the Andes Mountains when
they found a small valley, with no other animals or humans. Pérez
noticed that the valley had what appeared to be a natural fountain,
surrounded by two peaks of rock and silver snow.
```

OpenAI GPT-3 (figure from GPT-2). Brown et al. 2020.

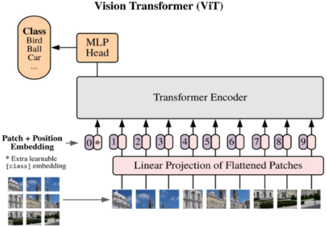
Transformer architectures become ubiquitous

Training these models on very large amounts of data -> rise of "foundation models"

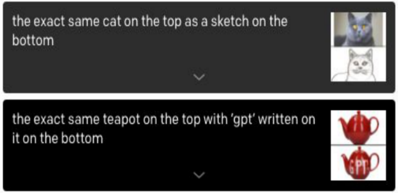
Transformer architecture can be applied to images as well!

Key idea: Convert image into sequence of patches (i.e. image tokens).

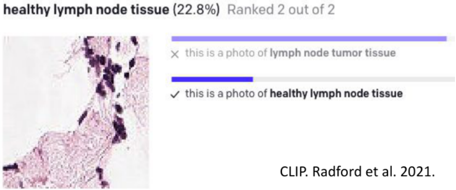
Can then benefit from Transformer architecture and its self-attention mechanism. Allows process of token encoding to incorporate contextual understanding (attention) from other tokens of the same (self) sequence.



ViT. Dosovitsky et al. 2021



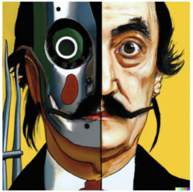
DALL-E. Ramesh et al. 2021.



CLIP. Radford et al. 2021.

2022: Major improvements in text-to-image generation models

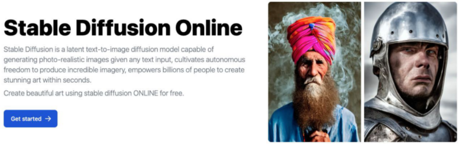
DALL-E 2
Ramesh et al. 2022.

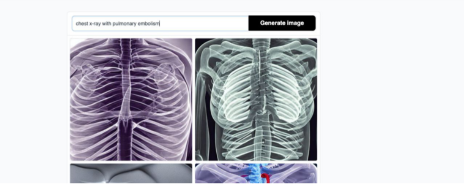


vibrant portrait painting of Salvador Dalí with a robotic half face

a shiba inu wearing a beret and black turtleneck

Meta Make-a-Video.
Singer et al. 2022





Stable Diffusion.
https://stablediffusionweb.com/

Late 2022: ChatGPT reaches public mainstream

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User help me write a short note to introduce myself to my neighbor

ChatGPT Hi there,

My name is [Your Name] and I just moved in next door. I wanted to reach out and introduce myself. I'm looking forward to getting to know you and the neighborhood. If you ever need anything or just want to chat, please don't hesitate to stop by or give me a call.

Best,
[Your Name]

User can you make it more formal?

ChatGPT Dear Neighbor,

I am writing to introduce myself. My name is [Your Name] and I have recently moved in next door. I wanted to take a moment to reach out and say hello. I am looking forward to getting to know you and the neighborhood.

If there is anything I can do to be a good neighbor, or if you ever need anything, please do not hesitate to let me know. I am happy to help in any way that I can.

Sincerely,
[Your Name]

OpenAI ChatGPT
https://openai.com/blog/chatgpt

Two major classes of vision foundation models

Representation Learners

Train on huge amounts of data

Objective: learn powerful generalist feature extractors that can be used for downstream computer vision tasks.

- Vision Only (e.g. DINOv2)
- Vision-Language (e.g. CLIP)

Generative Models

Train on huge amounts of data

Objective: learn to generate images and/or text.

- Text $\rightarrow$ Image (e.g. Stable Diffusion, DALL-E)
- Text, Image $\rightarrow$ Text (e.g. Flamingo, GPT4-V)

Detection Foundation Model: Grounding DINO

- Open-vocabulary detection with natural-language grounding.
- Uses transformer-based backbone for both image and text encoding.
- Matches object proposals in an image with user-provided text prompts (e.g. "a red backpack").
- Produces bounding boxes, labels, and confidence scores for detected objects.
- **Key architectural mechanisms**:
  - **Feature enhancer neck** with deep image-text cross-attention (not only late fusion).
  - **Language-guided query selection** to initialize decoder queries on relevant regions.
  - **Cross-modality decoder head** for iterative region-text refinement.
- **Takeaway:** strong transfer to unseen categories and prompts.

Grounding DINO: How It Works

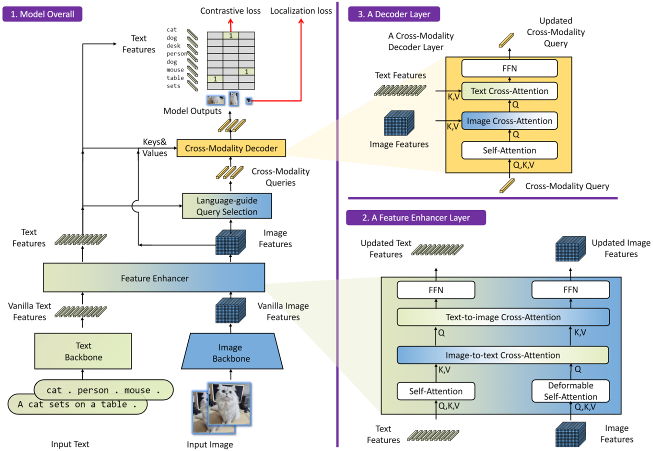
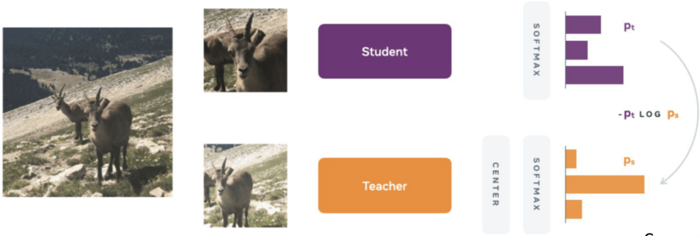


Figure 3. The framework of Grounding DINO. We present the overall framework, a feature enhancer layer, and a decoder layer in block 1, block 2, and block 3, respectively.

DINOv2: Transformer-based representation learner trained on 142 million images

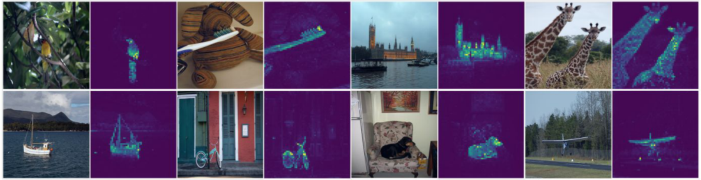
- DINO (self-Distillation with NO labels) is trained through self-supervised learning (no external label curation needed)
- Student-teacher self-supervised training objective: At a high level, images (teacher) should produce similar feature representations to randomly cropped portions of the image (student)



Caron et al. 2021, Oquab et al. 2023

DINOv2: Transformer-based representation learner trained on 142 million images

- Once trained, the DINO neural network backbone can be used as the “pre-trained” feature extractor for many downstream tasks

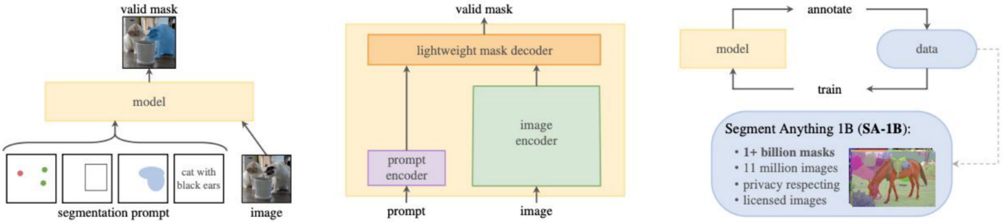


- DINOv2 made further implementation refinements and trained on 142 million images
- An example of a vision “**foundation model**”: a model trained on a very large amount of data that can be subsequently used or tuned for diverse tasks. DINOv2 was demonstrated to be a state-of-the-art backbone for many tasks.

Caron et al. 2021, Oquab et al. 2023

Segment Anything Model (SAM): A foundation model targeted for segmentation

- Foundation Model for promptable segmentation, based on a Transformer encoder-decoder architecture
- Trained on 1 billion masks from 11 million images, using the model in a data collection loop



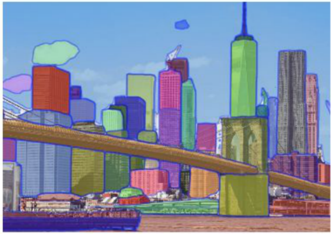
(a) **Task**: promptable segmentation (b) **Model**: Segment Anything Model (**SAM**) (c) **Data**: data engine (top) & dataset (bottom)

Kirrilov et al. 2023.

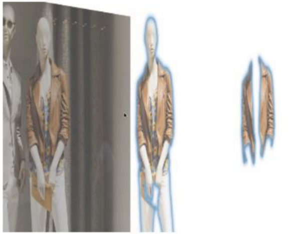
Segment Anything Model (SAM): A foundation model targeted for segmentation

- Allows segmentation based only from prompts, without needing to provide training examples of target classes (zero-shot segmentation)

Prompt it with interactive points and boxes.



Automatically segment everything in an image.



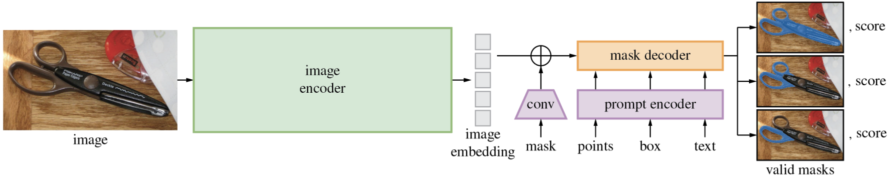
Generate multiple valid masks for ambiguous prompts.

Meta AI Segment Anything Model (SAM)
https://segment-anything.com/

Kirrilov et al. 2023.

How does SAM work?

- Encodes images into embeddings using a transformer backbone.
- User or downstream model provides prompts (points, boxes, text).
- The model predicts segmentation masks accordingly.
- Produces high-quality object masks across a wide range of settings without the need for fine-tuning.



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SAM 1

- Segments any object in any image.
- Requires as little as a single click to segment an object.
- Refine prediction with a few clicks.

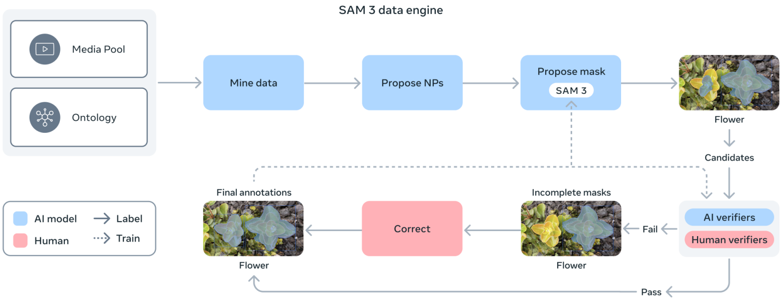
SAM 2

- Segments and tracks any object in any image or video.
- Uses click, box, or mask prompts.
- Tracks segment predictions across video frames.
- Refine prediction with a few clicks.

SAM 3

- Detect, segment and track every example of any object category in image or video.
- Uses text or examples as prompts.
- Segments and tracks objects from a click.
- Tracks segment predictions across frames.
- Detects and segments matching instances.
- Refine detection with visual examples.

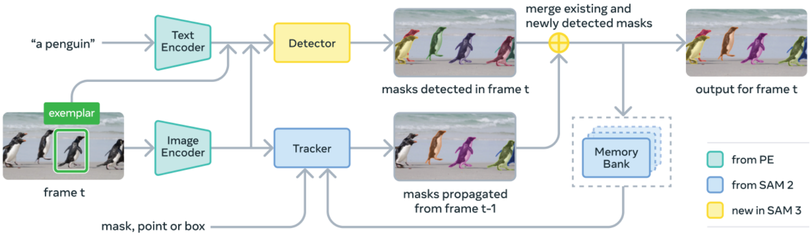
SAM 3 Architecture: Data Engine



SAM 3 data-engine loop from Meta blog post.

- Obtaining high-quality annotated images with segmentation masks and text labels across a broad range of categories and visual domains is a significant challenge. A scalable data engine combining SAM 3, human annotators, and AI models produced 4M+ high-quality, annotated concepts across diverse domains.
- SAM 3 uses a scalable data engine where AI models and annotators (human and Llama-based) collaborate in a feedback loop to rapidly mine, annotate, and verify segmentation datasets—significantly increasing throughput and data quality.

SAM 3 Architecture: Core Model



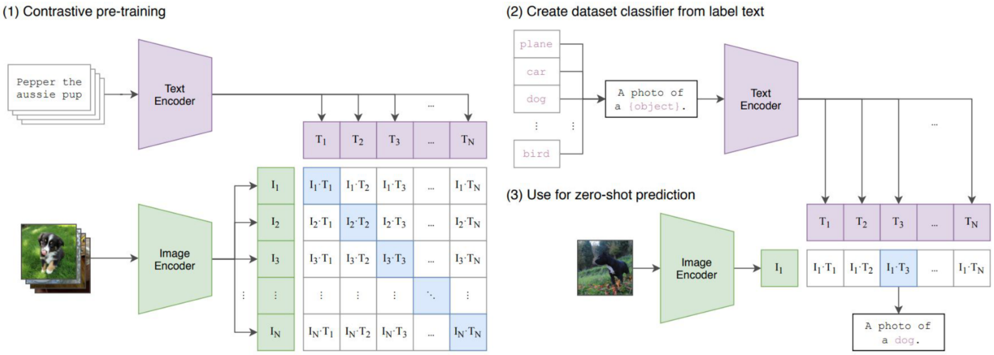
SAM 3 model summary and detector/tracker architecture.

Dual encoder-decoder transformer—serving as a detector for image-level capabilities—used together with a tracker and memory for video. The detector and tracker both receive vision-language features from an aligned Perception Encoder (PE) backbone.

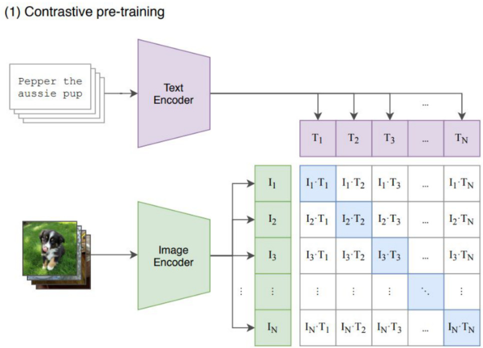
SAM 3.1 Inference Architecture: Object Multiplexing

SAM 3.1 multiplexed inference versus per-object computation.

- Process multiple tracked objects in one forward pass (*multiplexing*) instead of one pass per object.
- Shared object-level context reduces redundant compute and helps crowded-scene tracking.



Radford et al. 2021.

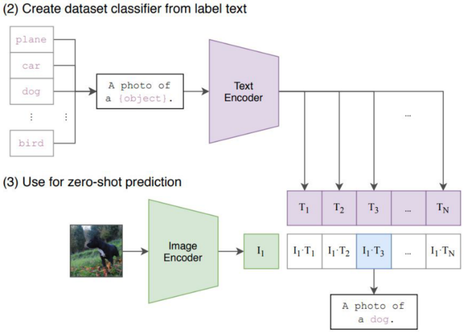


Radford et al. 2021.

Learn text and image encoders (i.e. feature extractors) through a self-supervised "contrastive" objective: images and their matching text captions should produce more similar features than non-matching ones

Trained on 400M image-text pairs: generated by searching for image-text pairs on the web, where text comes from a base query list of 500,000 queries comprising all words occurring at least 100 times in the English version of Wikipedia

Due to the contrastive training objective, image and text encoders output features in a shared representation space and the trained CLIP model can be used for zero-shot classification, i.e., perform N-way classification without showing the model paired examples of (input, class) for any of the N classes.

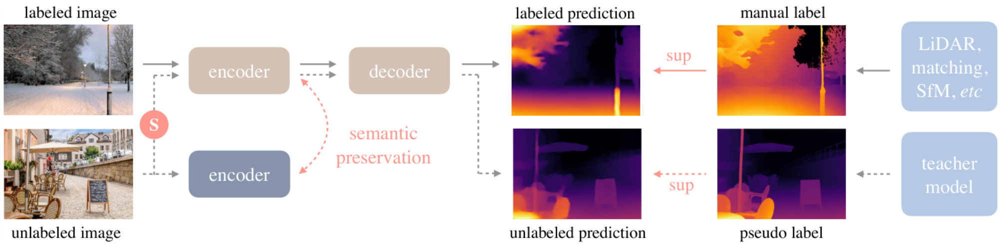


Radford et al. 2021.

Depth Estimation Foundation Model: Depth Anything

- Foundation model for single image depth estimation.
- Trained with a large-scale teacher-student pipeline:
  - $\sim$1.5M labeled images + $\sim$62M unlabeled images
- Encoder priors from DINOv2 improve cross-domain geometric understanding.
- Robust transfer across indoor/outdoor domains with limited task-specific supervision.
- Applications:
  - Robotics and autonomous driving
  - Virtual/augmented reality
  - 3D scene understanding and reconstruction

- **Teacher–student pipeline** on labeled depth plus large-scale unlabeled RGB: pseudo-labels from the teacher.
- DINOv2-style encoder + prediction head maps RGB to per-pixel depth.
- Outputs dense depth maps that transfer to segmentation and other geometric downstream tasks.

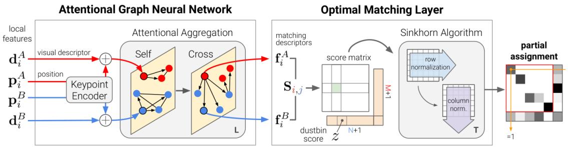


Yang et al., "Depth Anything," CVPR 2024 (OpenAccess).

SuperGlue: Foundation model for Keypoint Detection

- Combines graph neural networks and attention to find matches between sets of keypoints in different images.
- Generalizes across scenes and keypoint detector types.
- Applications: Visual localization and mapping, 3D reconstruction, Object pose estimation.

- Two main components: an attentional graph neural network (GNN) and an optimal matching layer.
- GNN uses keypoint encoder to combine each keypoint's position and visual descriptor into single vector.
- Alternating self-attention and cross-attention layers create powerful feature representations.
- The optimal matching layer builds an $M \times N$ score matrix between keypoints of both images, appends additional rows/columns ("dustbins") for potential unmatched keypoints.
- Uses Sinkhorn algorithm for T iterations to find optimal partial matching assignment.

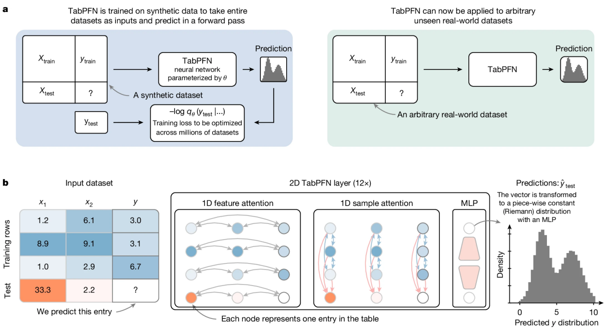


Sarlin et al., "SuperGlue: Learning Feature Matching with Graph Neural Networks," CVPR 2020 (OpenAccess).

TabPFN: Tabular Pretrained Foundation Network

- Trained to "meta-learn" classification and regression solutions for tabular datasets.
- One forward pass – inference in less than 1 second.
- Zero-shot predictions for new, small tabular datasets.
- Applications: Science data analysis, health records, finance.

TabPFN: Architecture and Capabilities



Hüser et al., "TabPFN: Pretrained Transformers for Tabular Problems," ICML 2023 (arXiv:2207.01848)

- Learns a prior over tabular tasks by pretraining on large synthetic task distributions.
- Performs in-context learning on a new dataset in one forward pass.
- Two-way attention captures dependencies across both rows and features.

- Model of brain-response dynamics.
- Frozen encoders for video/audio/text + transformer integration backbone.
- Supports zero-shot transfer to unseen subjects/tasks at population level.

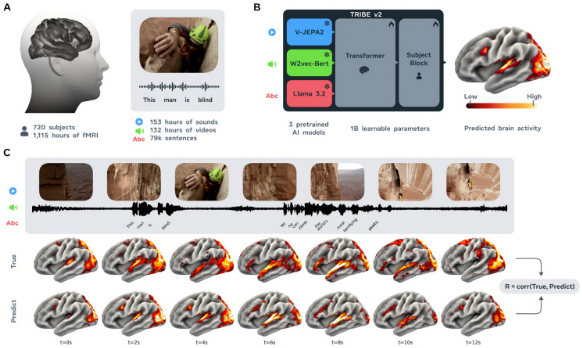


Figure from TRIBE v2 paper: data $\rightarrow$ pretrained embeddings $\rightarrow$ predicted cortical response vs. measured mean response

TRIBE v2: Encoding Performance (Results)

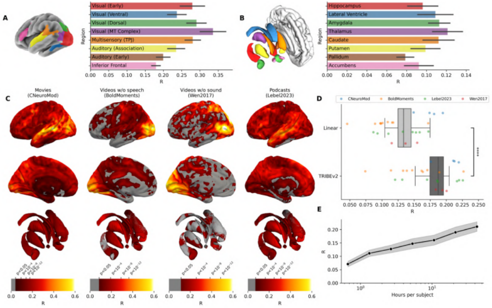


Figure from TRIBE v2 paper: Cortical/subcortical encoding scores; vs. linear baseline; scaling with training data.

- **Accurate whole-brain prediction:** TRIBE v2 accurately predicts fMRI responses across the entire brain.
- **Cortical encoding scores:** Achieves high encoding scores in key cortical regions.
- **Subcortical encoding performance:** Robust encoding in 8 subcortical regions.
- **Per-subject overview:** Individual subject encoding scores, with dots for TRIBE v2.
- **Scaling laws:** Demonstrates average encoding scores on the Algonauts dataset improve as more training data hours per participant are used.

Video Foundation Model: InternVideo

- Large-scale pre-trained video foundation model.
- Developed to generalize across diverse video understanding tasks by learning rich spatio-temporal features adaptable to unseen video domains.
- Capabilities: Recognition, localization, captioning, segmentation, search, QA, and multi-modal video reasoning.

- Stage 1: Spatiotemporal perception (masked reconstruction)
- Stage 2: Semantic alignment (video-text/audio contrastive learning)
- Stage 3: World-modeling style reasoning (next-token objectives)

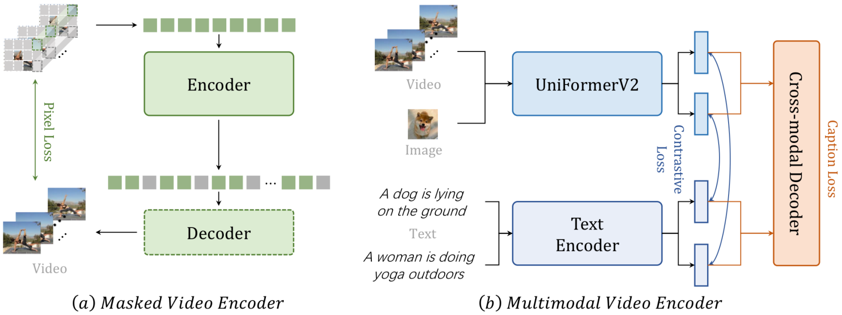


Figure from InternVideo paper: InternVideo2 architecture.

Summary and Discussion

- Foundation models are reshaping AI across all domains (vision, language, tabular, medicine, video).
- Key trends: Scalability, data diversity, task generalization, multi-modal grounding.
- Challenges: Compute costs, ethical issues, responsible deployment, domain adaptation.
- **Discussion:**
  - How will foundation models change your research?
  - What domains remain out-of-reach for today's foundation models?

References

1. Bommasani et al., "On the Opportunities and Risks of Foundation Models." arXiv:2108.07258.
2. Liu et al., "Grounding DINO: Marrying DINO with Grounded Pre-Training for Open-Set Object Detection." arXiv:2303.05499.
3. Kirillov et al., "Segment Anything." arXiv:2304.02643.
4. "SAM 3: Segment Anything with Concepts." Meta AI + arXiv:2511.16719 + OpenReview.
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